Problem 2: Exploring Edge Impulse (20 points)

1. [Dis] (5 pts) What types of "Learning Blocks" are available under the free-tier Edge Impulse account (e.g., supervised learning—classification and regression, unsupervised learning—clustering, self-supervised learning, etc.)?
For each type, provide at least one example model that can be implemented using the free tier.

Classification (Keras)
Example: Classifying hand gestures using accelerometer and gyroscope data from a wearable device.

Regression (Keras)
Example: Predicting room temperature from environmental sensor readings.

Anomaly Detection (K-means)
Example: Detecting abnormal vibration patterns in a machine motor.

Anomaly Detection (GMM)
Example: Identifying unusual energy consumption behavior in a smart home system.

Visual Anomaly Detection (FOMO-AD)
Example: Detecting defective products on a conveyor line.

Image Classification (using Transfer Learning)
Example: Identifying different species of plants from camera images.

Keyword Spotting (using Transfer Learning)
Example: Recognizing spoken commands like "start" or "stop" for a smart device.

Object Detection (using MobileNetV2 SSD FPN)
Example: Detecting cars, pedestrians, and bicycles in traffic camera footage.

Object Detection (using FOMO)
Example: Counting pieces of fruit on a tree using an embedded camera.

Classical ML
Example: Simple motion classification using extracted IMU features and a lightweight ML model.
https://docs.edgeimpulse.com/studio/projects/learning-blocks

2. [Dis] (5 pts) List all the types of layers that can be used under the "Classifier" section in the free-tier Edge Impulse. Briefly explain the purpose of each layer identified.

Dense
Fully connected layer, the simplest form of a neural network layer. Use this for processed data, such as the output of a spectral analysis DSP block.
1D Convolution / pooling
Learn features that take spatial information into account along a single dimension. Use this for raw data, or for DSP blocks that output spatial data, such as the MFCC block.
2D Convolution / pooling
Learn features that take spatial information into account along two dimensions. Use this for raw data, or for DSP blocks that output spatial data, such as the MFCC block.
Reshape
Turn one-dimensional data from a DSP block into multi-dimensional data. Use this as an input to a convolutional layer. Use this for deep learning on raw data, or to process MFCC output.
Flatten
Flatten multi-dimensional data into a single dimension. You need to flatten data from a convolutional layer before returning.
Dropout
Reduce the risk of a model overfitting your dataset by randomly cutting a fraction of network connections during training. Can be helpful if your model's training performance is better than its validation performance.

(pulled direct from clicking 'add an extra layer')

3. [Dis] (5 pts) What types of model compression options are available under the "Deployment" section in the free-tier Edge Impulse? List all available methods.

I think just this for us:
Quantized (INT8) models
Unoptimized Float32 models
EON Compiler
EON Compiler (RAM optimized)? maybe?
Tensorflow lite

4. [Dis] (5 pts) What input modalities are supported for synthetic data generation in the free-tier version of Edge Impulse? For each modality, list the synthetic data generation methods/algorithms provided. Briefly explain why synthetic data is important in machine learning applications.

Image Generation
Flux-Pro Synthetic Image Generator
    Uses Black Forest Labs' Flux-Pro model through the Replicate API.
    Generates realistic synthetic images from text prompts.
    Useful for expanding image classification or object detection datasets.

DALL-E 3 Synthetic Image Generator

Uses OpenAI's DALL-E 3 model.

Generates custom images from natural language descriptions.

Useful when collecting real images is difficult or expensive.

Audio / Voice Generation

Whisper Synthetic Voice Generator

Uses OpenAI text-to-speech models.

Generates synthetic spoken keywords or phrases.

Useful for keyword spotting and speech recognition systems.

Simulation-Based Synthetic Data

NVIDIA Omniverse

Creates simulated environments and synthetic sensor/camera data.

Often used for robotics, autonomous systems, and object detection training.